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# Workshop on Multinomial- Processing-Tree (MPT) Modelling

## Basics of MPT Modeling



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# Learning Goals

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## What you should learn in this workshop:

- How can we **measure latent psychological processes**?
- What are the **statistical foundations** of MPT modeling?
- How to **develop and apply** MPT models?
- How to use **Bayesian hierarchical modeling**?

## Why is this useful?

- General understanding of **cognitive modeling**: Connecting psychological theory & data analysis
- MPT models are applied in **many areas**
  - memory, reasoning, social cognition, implicit attitudes, evaluative conditioning, decision making, aging, dishonesty, eyewitness detection, environmental psychology, ...

# Workshop

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## Who are you?

- Your name & background
- Why MPT modeling?

## Course requirements

- Reading of the literature
- Active participation
- **Exercises**: Hand in solutions as PDF via email
- Responses may be given in **bullet point style**
- No grades, just **pass/fail**

# Basics of MPT Modeling

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## Overview

1. Measuring cognitive processes
2. Formal model structure
3. Identifiability
4. Parameter estimation
5. Model assessment & comparison

# Data Structure

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- Required type of data: Multinomial models are tailored to **discrete, categorical data**
    - yes/no responses
    - correct/incorrect judgments
    - multiple choice tasks
    - number or type of recalled words
    - ...
  - **Psychological data** are typically discrete in nature
    - If not, they can often be transformed into discrete data
    - Response time bins, rankings of numerical judgments, ...
- many psychological paradigms generate frequency data appropriate for MPT modeling.

# Measuring Cognitive Processes

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## MPT models...

- ... provide **explanations** of observed frequency data in terms of basic parameters with clear-cut psychological interpretations;
- ... these **parameters** represent probabilities of latent psychological processes (or latent psychological states) underlying human behavior;
- ... in other words, these models **disentangle and measure** the contributions of different psychological processes to frequencies of observable behaviors.

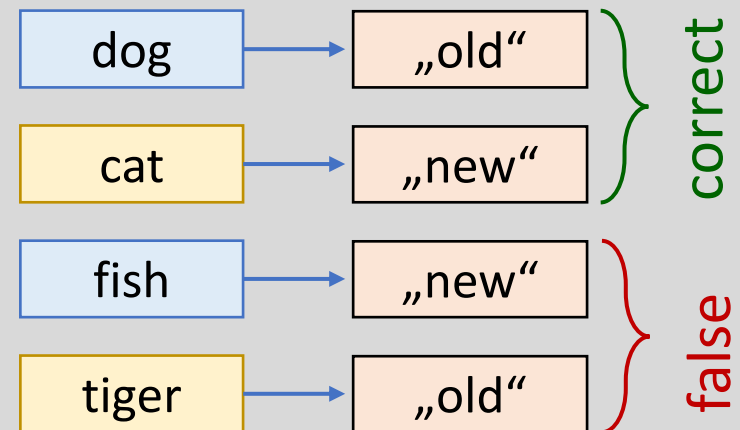
# Example: Recognition Memory

- Paradigm:
  - Old-new recognition memory test
- Two Conditions:
  - Old items (targets)
  - New items (lures)
- Categorical (dichotomous) dependent variable:
  - „old“ vs. „new“ judgment

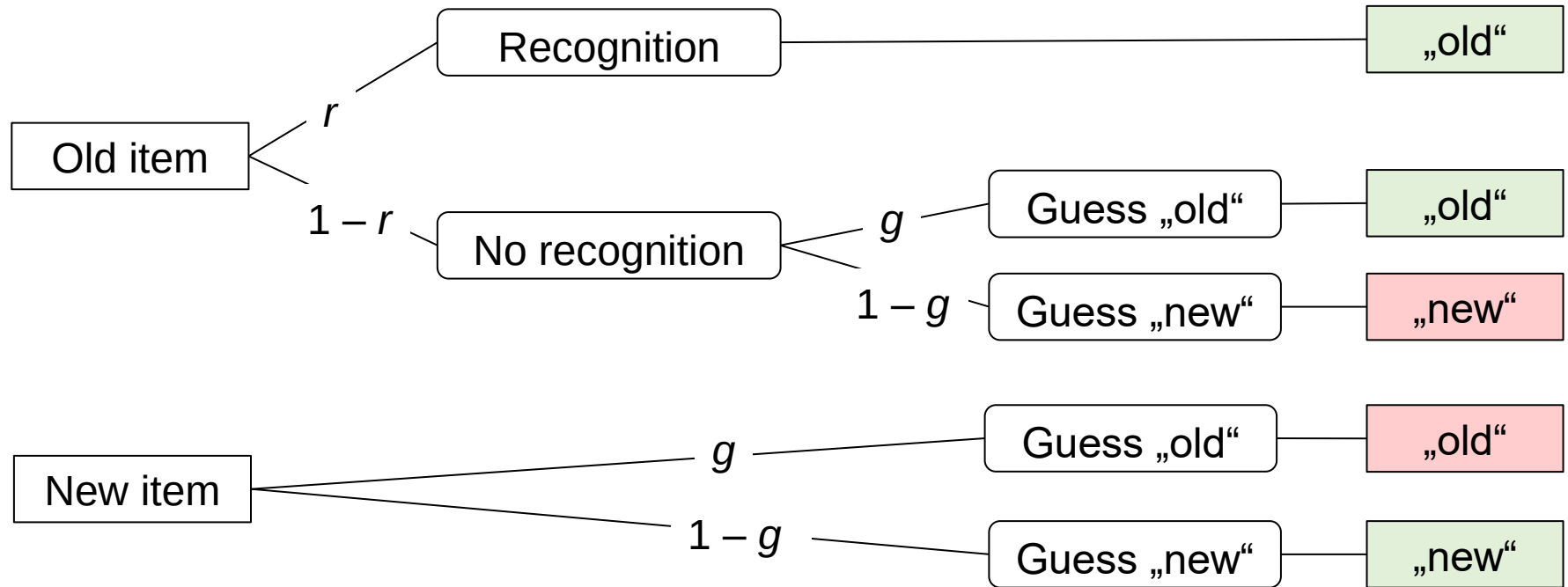
## 1. Study phase



## 2. Test phase



# 1-High Threshold Model (Blackwell, 1963)



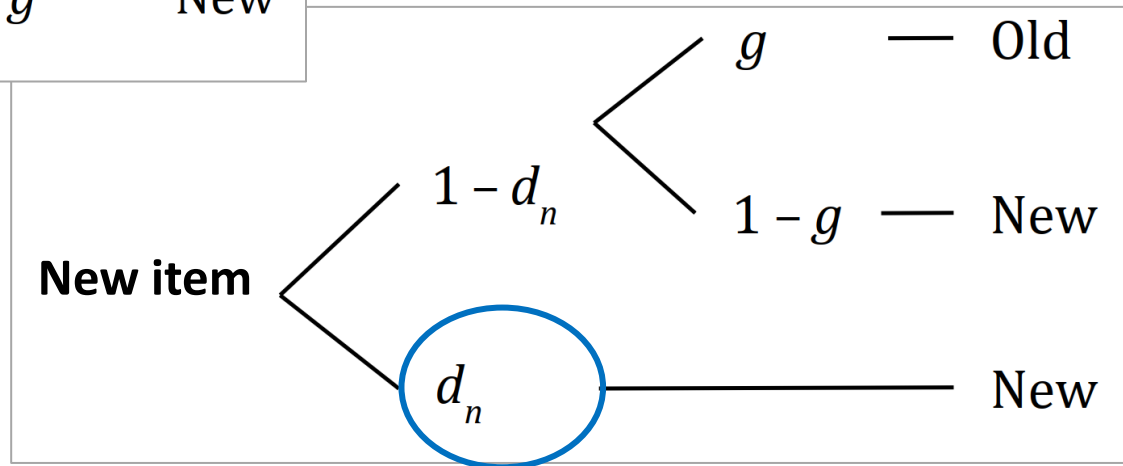
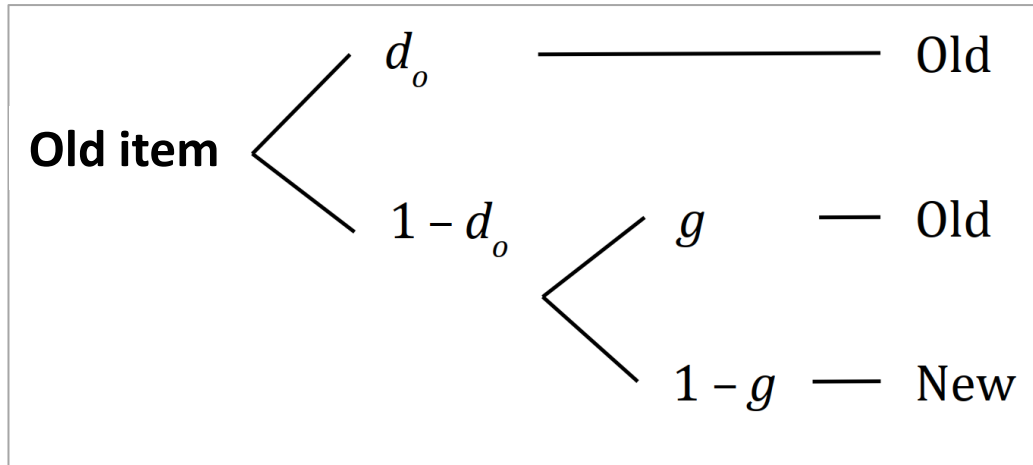
## Model equations:

$$P(\text{„old“} \mid \text{old item}) = r + (1 - r) \cdot g$$

$$P(\text{„old“} \mid \text{new item}) = g$$



# 2-High Threshold Model



**Model equations:**

$$P(\text{„old“} \mid \text{old item}) = d_o + (1 - d_o) \cdot g$$

$$P(\text{„old“} \mid \text{new item}) = (1 - d_n) \cdot g$$

# Terminology

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- **“Multinomial”**

- MPT models assume that observations are sampled independently from one or more **multinomial distributions**
- The frequency data structure can be univariate or multivariate

- **“Processing”**

- Assumption that a **finite number of latent processes** generate the observed responses
- Goal: Drawing inferences about these processes (e.g., via parameter estimation or hypothesis testing)

- **“Tree”**

- Models can be depicted as **probability trees**

# Pair-Clustering Model

- How can we obtain separate measures for **storage and retrieval** of information in long-term memory?
- Empirical Paradigm: **Free recall** of word list consisting of
  - a) Semantically related **word pairs** („chair“ & „table“)
  - b) Unrelated **singletons** („rose“ & no other flowers in study list)

## 1.) Study list

1. chair
2. lion
3. rose
4. table
5. hat
6. tiger

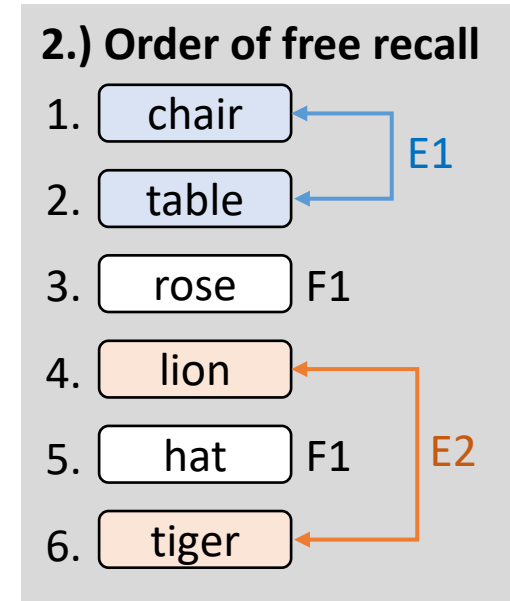
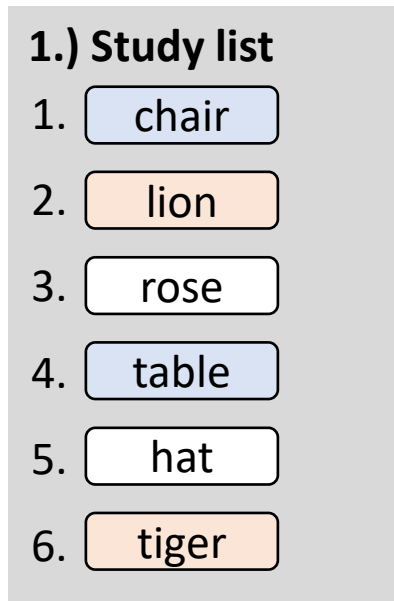
## 2.) Free recall of words

1. chair
2. rose
3. table
4. lion
5. ...

**Not recalled:** hat tiger ...

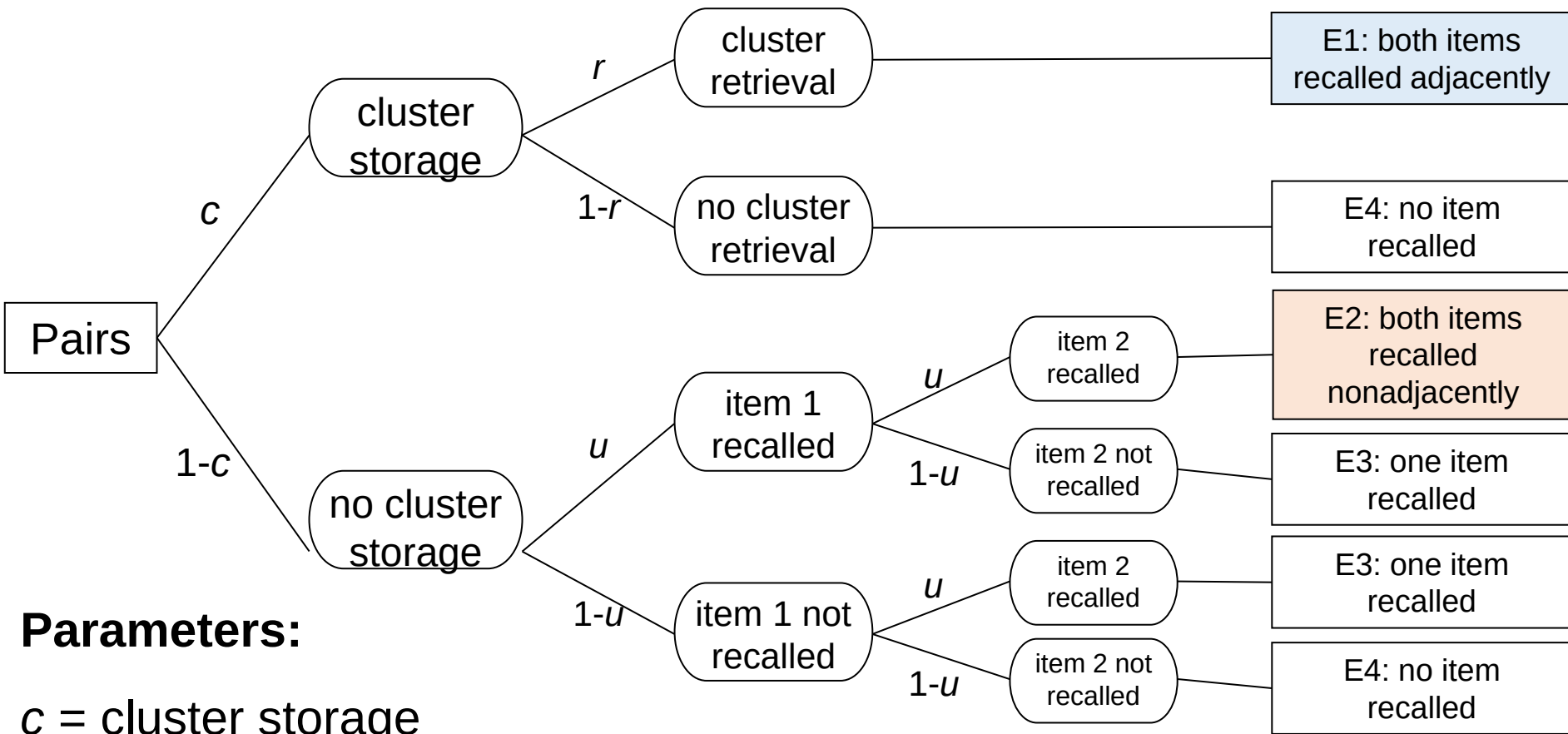
# Pair-Clustering Model

- The system of response categories for the MPT model is provided by using a **specific scoring of responses**:
- For **word pairs**:
  - E1 = both words recalled *adjacently*
  - E2 = both words recalled *non-adjacently*
  - E3 = one word recalled
  - E4 = no word recalled
- For **singletons**:
  - F1 = Recalled
  - F2 = Not recalled



# Pair-Clustering Model

(Batchelder & Riefer, 1980, 1986)

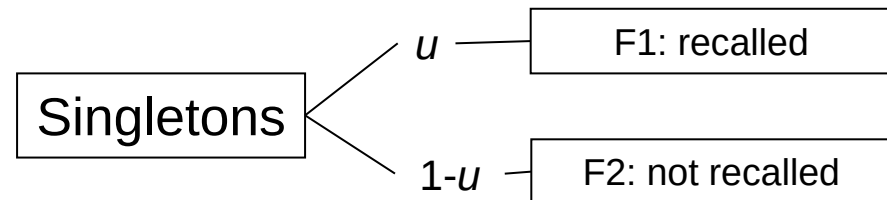


## Parameters:

$c$  = cluster storage

$r$  = cluster retrieval

$u$  = storage & retrieval of singletons



# Core Features of MPT Models

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## MPT models...

- ... **explain** discrete, categorical data (frequencies)
- ... are **tailored** to specific paradigms & theories
  - Benefit: Psychological theories are specified more precisely.
- ... can be **applied** in various ways
  - Measure the probability of certain psychological processes
  - Test the effect of experimental manipulations
  - Compare different groups of participants
  - Test different theoretical accounts (= different models)

# Outlook

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## **Research questions**

1. Model development
2. Model validation
3. Model application

## **Statistical aspects**

- Define categories & model equations
- Check model identifiability
- Optimize statistical power of model tests
- Assess goodness-of-fit
- Perform model selection

# Basics of MPT Modeling

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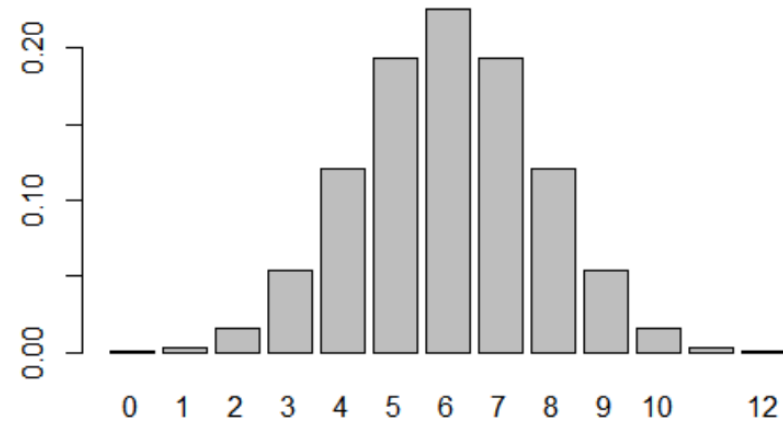


# Binomial Models

## Binomial model

- One variable with 2 categories
- Data: Observed response frequencies  $\mathbf{n} = (n_1, n_2)$
- Parameters: Vector of category probabilities  $\mathbf{p} = (p_1, p_2)$
- Given independent sampling, the frequencies follow a binomial distribution:

$$p(n_1, n_2) = \binom{N}{n_1} p_1^{n_1} (1 - p_1)^{N - n_1} = \frac{N!}{n_1! n_2!} p_1^{n_1} p_2^{n_2}$$



### Binomial Distribution

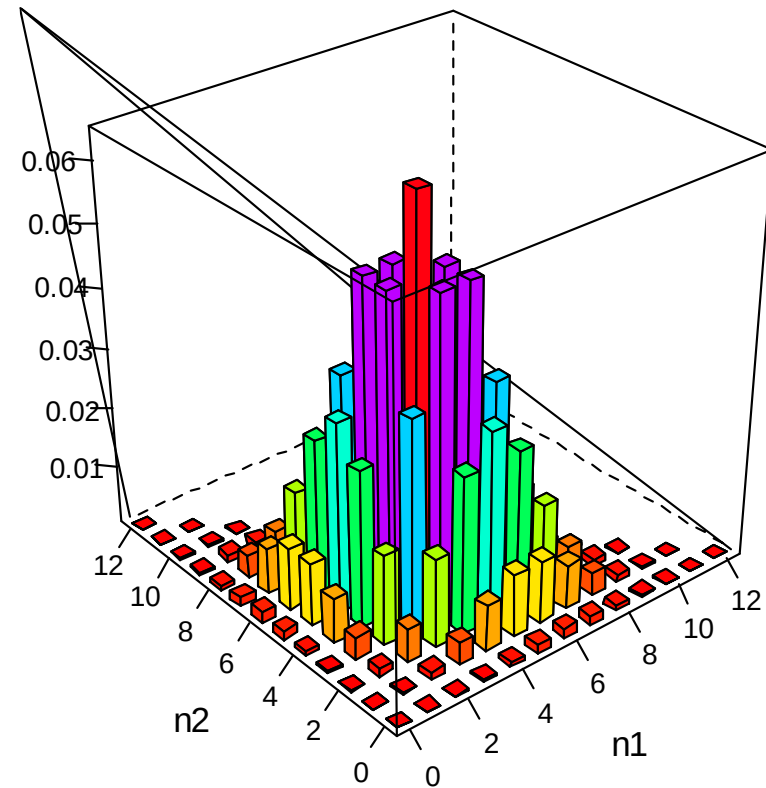
- $N = 12$  responses
- $p_1 = p_2 = 1/2$

# Multinomial Models

## Multinomial model

- One variable with  $J$  categories
- Data: Observed response frequencies  $\mathbf{n} = (n_1, n_2, \dots, n_J)$
- Parameters: Vector of category probabilities  $\mathbf{p} = (p_1, p_2, \dots, p_J)$
- Given independent sampling, the frequencies follow a **multinomial distribution**:

$$p(n_1, n_2, \dots, n_J) = \frac{N!}{n_1! n_2! \dots n_J!} p_1^{n_1} p_2^{n_2} \dots p_J^{n_J}$$



## Multinomial Distribution

- $N = 12$  responses
- $J = 3$  categories
- $p_1 = p_2 = p_3 = 1/3$

# Parameterized Multinomial Models

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## Parameterized Multinomial Model

- The category probabilities  $\mathbf{p} = (p_1, p_2, \dots, p_J)$  are rewritten as functions of the **latent parameters**  $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_S)$
- Based on the simple multinomial model, we define a set of **model equations**  $f(\boldsymbol{\theta})$ :
  - $p_1 = f_1(\theta_1, \theta_2, \dots, \theta_S)$
  - $p_2 = f_2(\theta_1, \theta_2, \dots, \theta_S)$
  - ....
  - $p_J = f_J(\theta_1, \theta_2, \dots, \theta_S)$
- The set of possible values of  $S$  latent parameters  $\theta_s$  is called **parameter space**  $\Omega$  of the model.

# Parameterized Multinomial Models

## Example

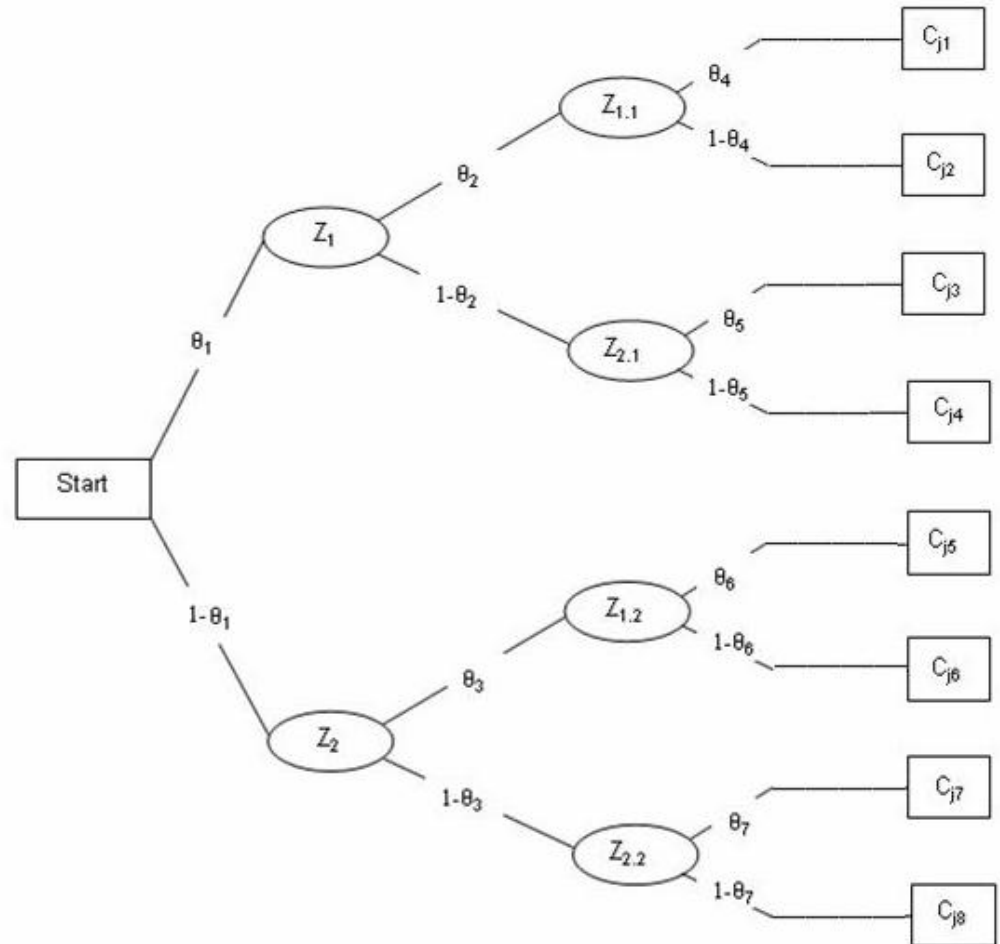
- Response categories in **old-new recognition memory**:
  1. *hit* = correct “old” response to old items
  2. *miss* = incorrect “new” response to old items
- Model equations of the **1-high threshold model (1HTM)** of recognition memory

$$\begin{aligned} p(\textit{hit}) &= p(\text{“old”} \mid \text{old item}) = r + (1 - r) g \\ p(\textit{miss}) &= p(\text{“new”} \mid \text{old item}) = (1 - r) (1 - g) \end{aligned}$$

$$p(n_{\textit{hit}}, n_{\textit{miss}} \mid r, g) = \frac{N!}{n_{\textit{hit}}! n_{\textit{miss}}!} [r + (1 - r) g]^{n_{\textit{hit}}} [(1 - r) (1 - g)]^{n_{\textit{miss}}}$$

# MPT Models

- What distinguishes **MPT models** from other multinomial models?
- MPT models assume a **specific form** of the model equations
- Branch probabilities of a **binary probability tree**



# Formal Definition of MPT Models

## MPT models

- A specific class of **parameterized** multinomial models
- Each parameter  $\theta_s$  is in the interval  $[0, 1]$  (= a probability)
- The **structure** of the model equations is given as:

$$p_j = \sum_{i=1}^{I(j)} c_{ij} \prod_{s=1}^S \theta_s^{a_{ijs}} \cdot (1 - \theta_s)^{b_{ijs}}, \quad \sum_{j=1}^J p_j = 1, \quad \theta_s \in [0, 1]$$

where  $s$ : Parameter index

$j$ : Category index

$i$ : Branch index

$c_{ij}$ : positive real number

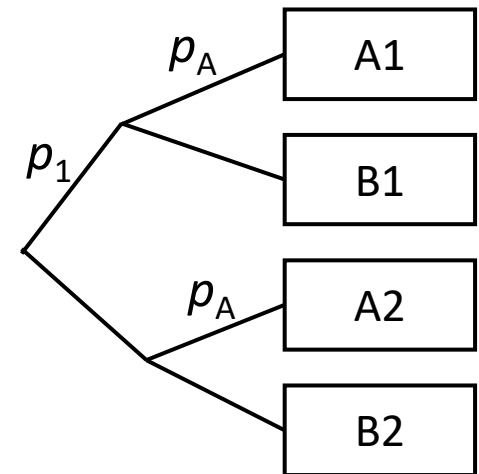
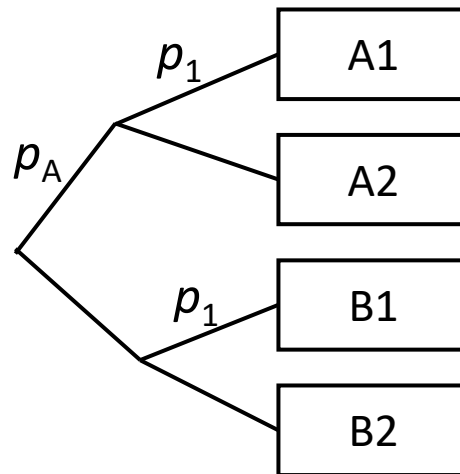
$a_{ijs}, b_{ijs}$ : nonnegative integer number (often 0 or 1)

# Uniqueness of the „Tree“

- A **binary probabilistic event tree** uniquely determines a system of MPT model equations.
- However: it is *not* true that any **system of MPT model equations** uniquely determines a specific tree diagram.

## Counter Example 1

- Level switching in **independence models**
  - $p(A1) = p_A p_1$
  - $p(A2) = p_A (1 - p_1)$
  - $p(B1) = (1 - p_A) p_1$
  - $p(B2) = (1 - p_A) (1 - p_1)$



# Counter Example 2

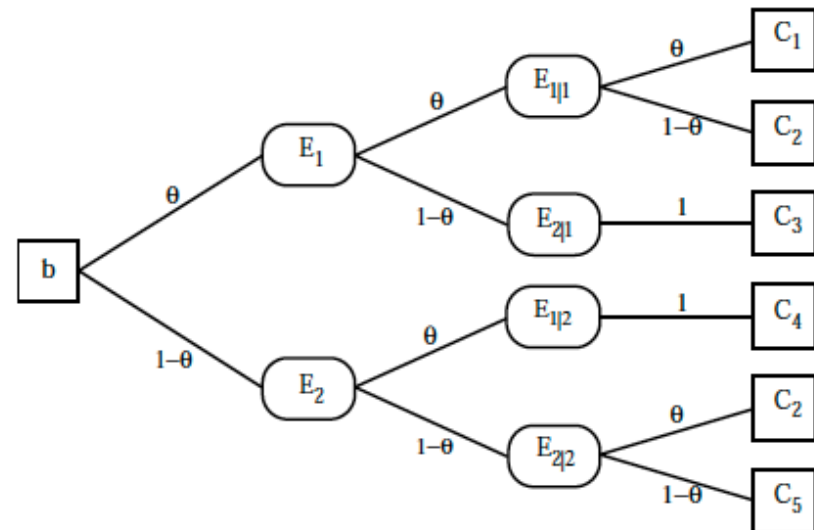
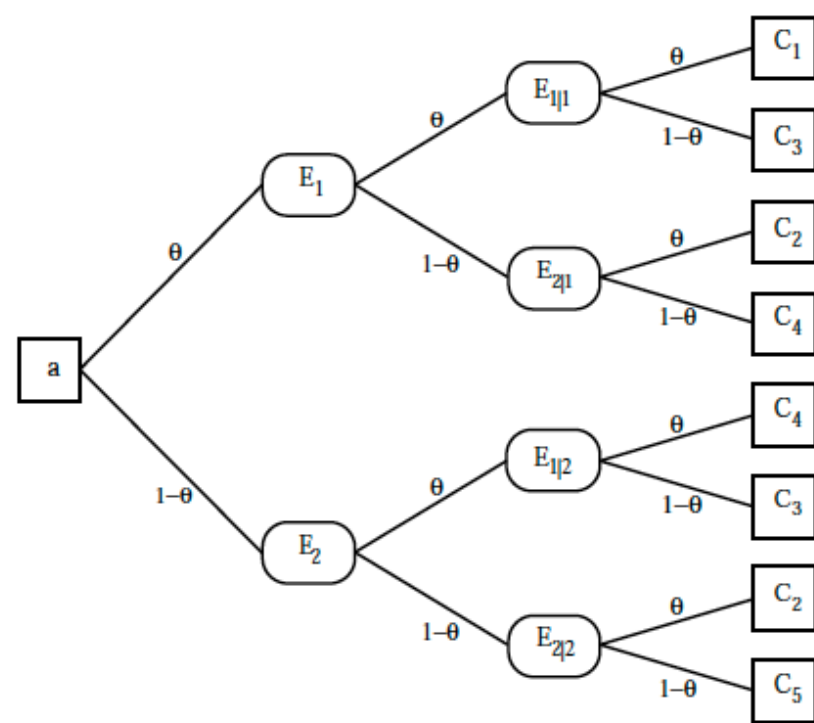
$$p_1(\theta) = \theta^3$$

$$p_2(\theta) = \theta \cdot (1 - \theta)$$

$$p_3(\theta) = \theta \cdot (1 - \theta)$$

$$p_4(\theta) = \theta \cdot (1 - \theta)$$

$$p_5(\theta) = (1 - \theta)^3$$





# Basics of MPT Modeling

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# Identifiability

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- A MPT model defines a **mapping**  $f: \Omega \rightarrow \mathcal{P}$ 
  - **Parameter space**  $\Omega$  = the set of all possible parameter vectors  $\theta$
  - **Data space**  $\mathcal{P}$  (more precisely: space of category probabilities)  
= the set of all possible category probability vectors  $\mathbf{p}$

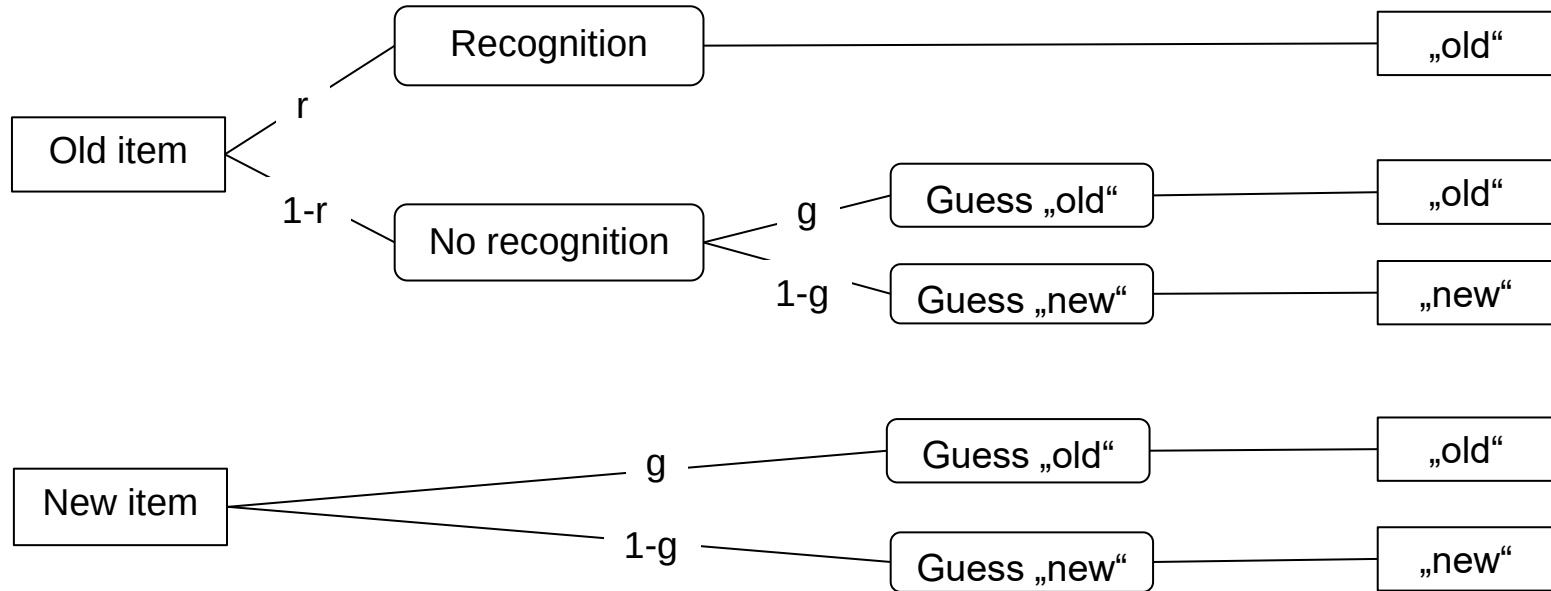
## Global identifiability

- A MPT model is globally identified if the mapping  $f$  is one-to-one **for all parameters**  $\theta$  in  $\Omega$ .

## Local identifiability

- A MPT model is locally identified if the mapping  $f$  is one-to-one in the **neighborhood of a specific point**  $\theta_0$  in  $\Omega$ .

# 1-High-Threshold Model (Blackwell, 1953)

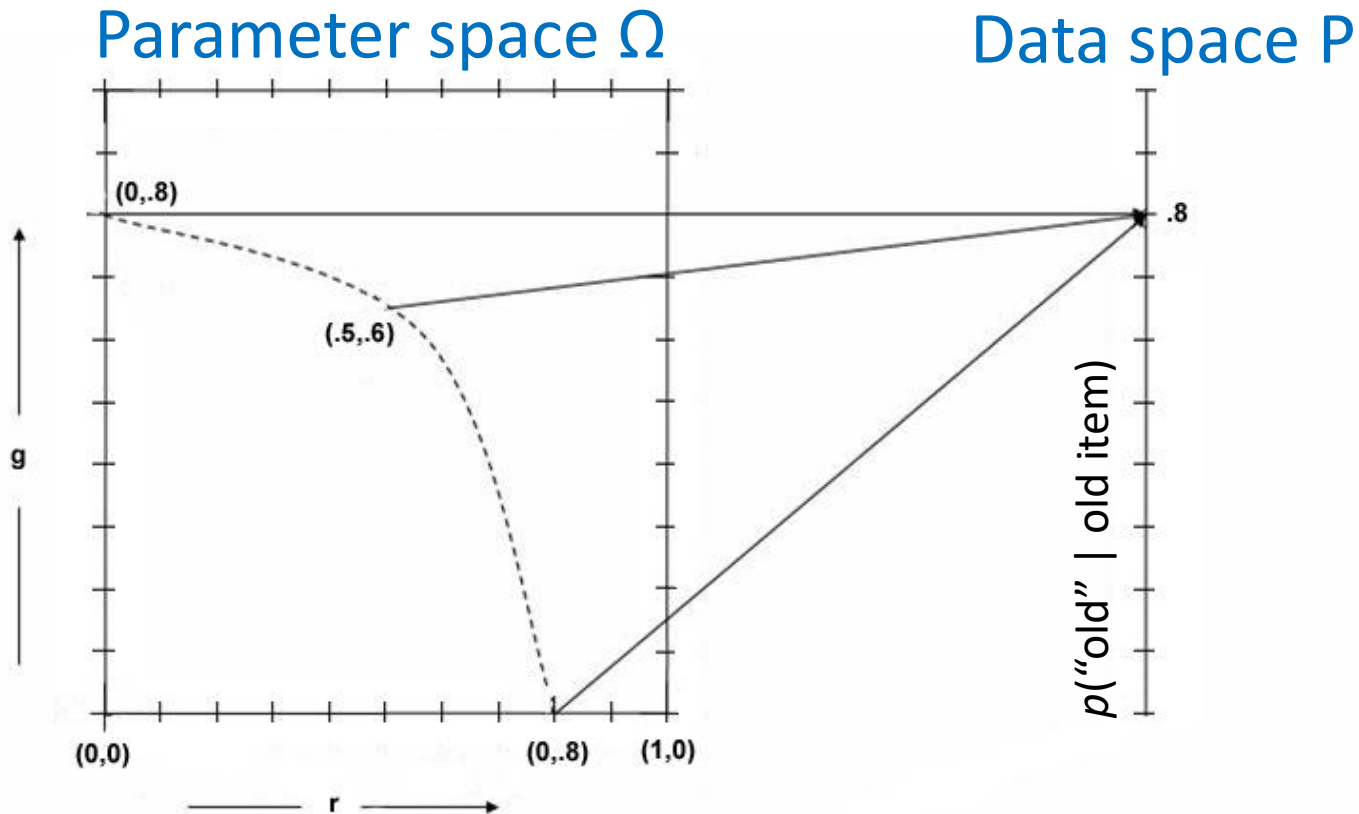


## Parameters

- $r$  = probability of **recognition**
- $g$  = probability of **guessing** „old“ given no recognition

# Example 1: Nonidentifiability

- 1-high-threshold model **limited to old items**:  
$$p(\text{"old"} \mid \text{old item}) = r + (1 - r) g$$
- The model is **not identified**



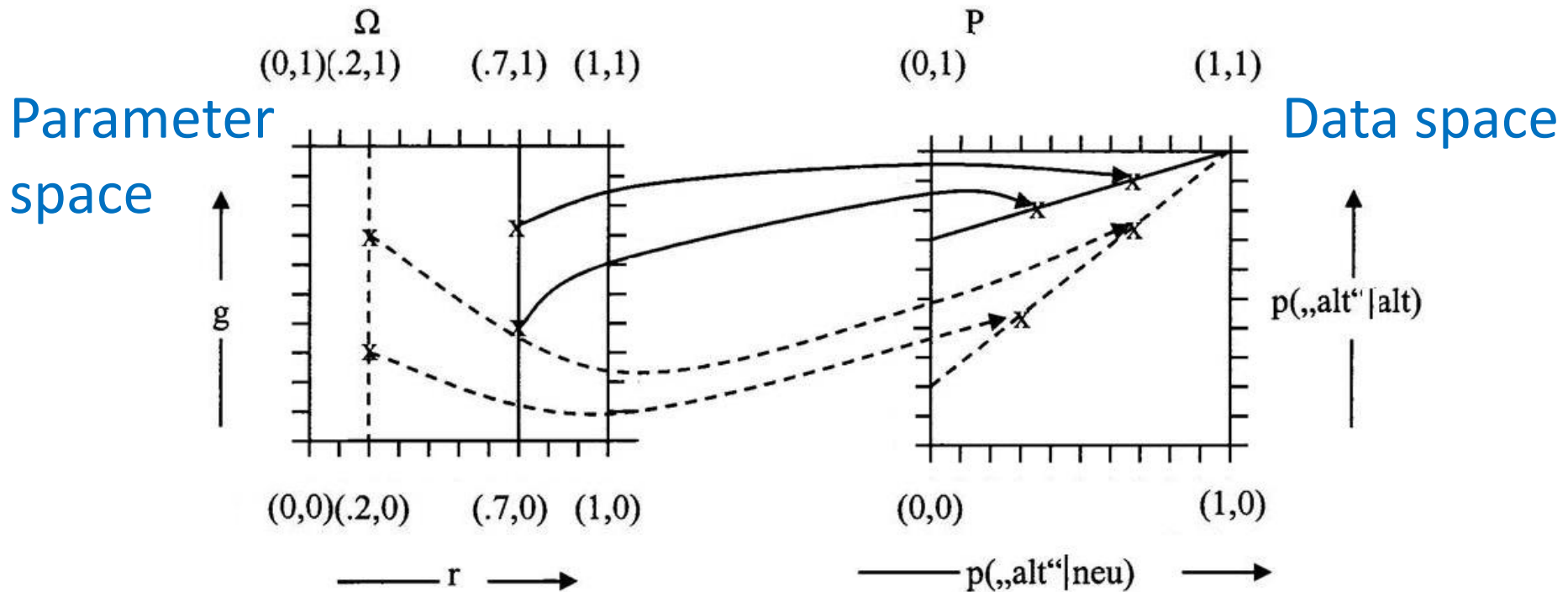
# Example 2: Identifiability

- 1-high-threshold model for old & new items

- $p(\text{"old"} \mid \text{old item}) = r + (1 - r) \cdot g$

- $p(\text{"old"} \mid \text{new item}) = g$

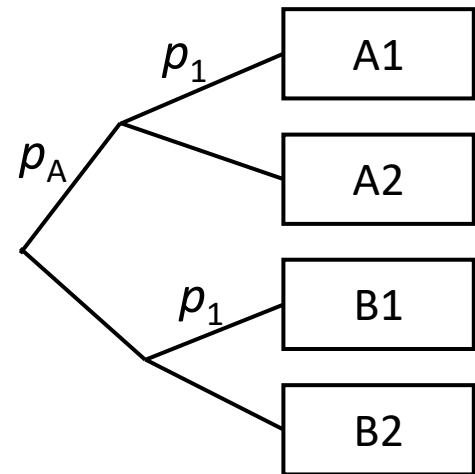
- The model is globally identified



# Identifiability: Two Important Theorems

## Observable branches

- A model is **always globally identified** if each of its branches terminates in a new empirical category (Hu & Batchelder, 1994).



## Number of parameters

- A model **cannot have more parameters than degrees of freedom** in the data.
- Hence, a necessary but not sufficient condition of identifiability for the number of parameters  $S$  is:

$$S \leq \sum_{k=1}^K (J_k - 1) \quad [J_k = \text{number of response categories in condition } k]$$

# Identifiability: Jacobian Matrix

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## Jacobian matrix

- Matrix of the **first partial derivatives** of all model equations with respect to all parameters  $\theta_s$
- **$r = \text{maximum rank}$**  of the Jacobian matrix across  $\Omega$  (can be computed in multiTree)
- General rules:
  - If  **$r < S$** , then the model is neither locally nor globally identified.
  - If  **$r = S$** , then the model is locally identified (but not necessarily globally).

# Remedies for Nonidentifiable Models

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## How to get an identifiable MPT model?

### a) Less free parameters → Parameter constraints

- Parameter fixations  $\theta_s = c$  (with  $c$  = constant number)
- Equality constraints  $\theta_s = \theta_t$  (for two parameters)

### b) Add more empirical categories

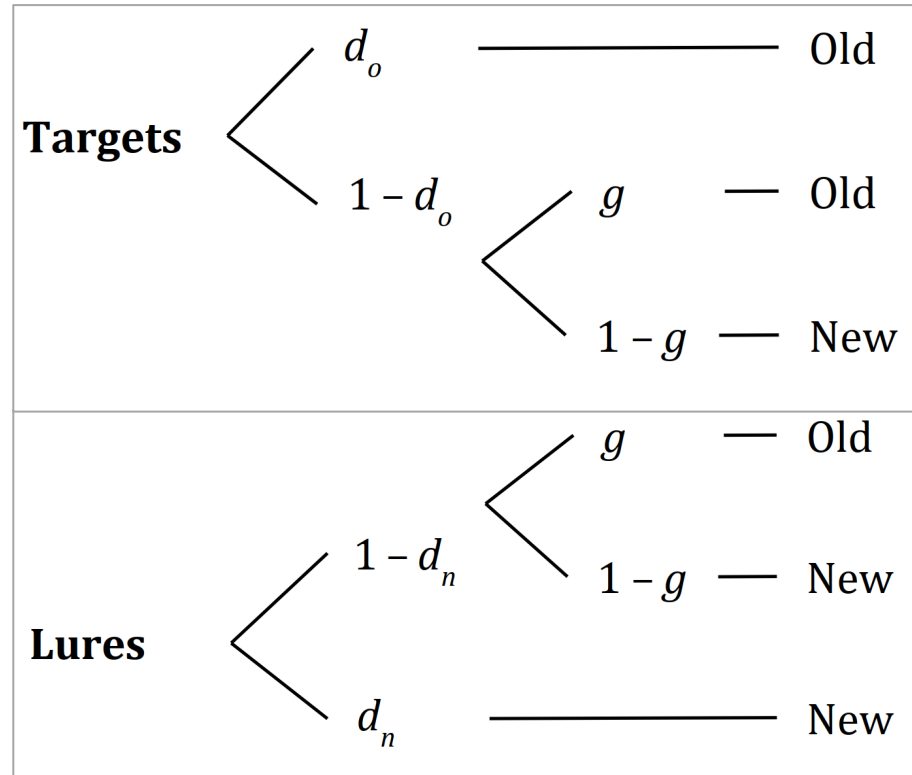
- Additional conditions with no (or few) additional parameters
- Selective manipulations of parameters



# Identifiability: Example

## 2-High Threshold Model

- Parameters:  $S = 3$  ( $d_o$ ,  $d_n$ ,  $g$ )
  - Free categories:  $df = 2$
- not identifiable!

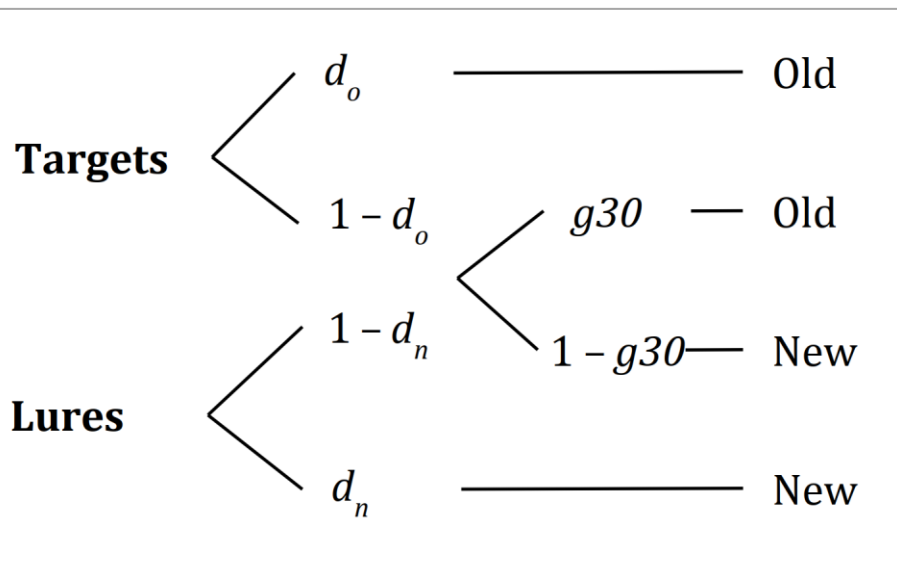


## Solutions

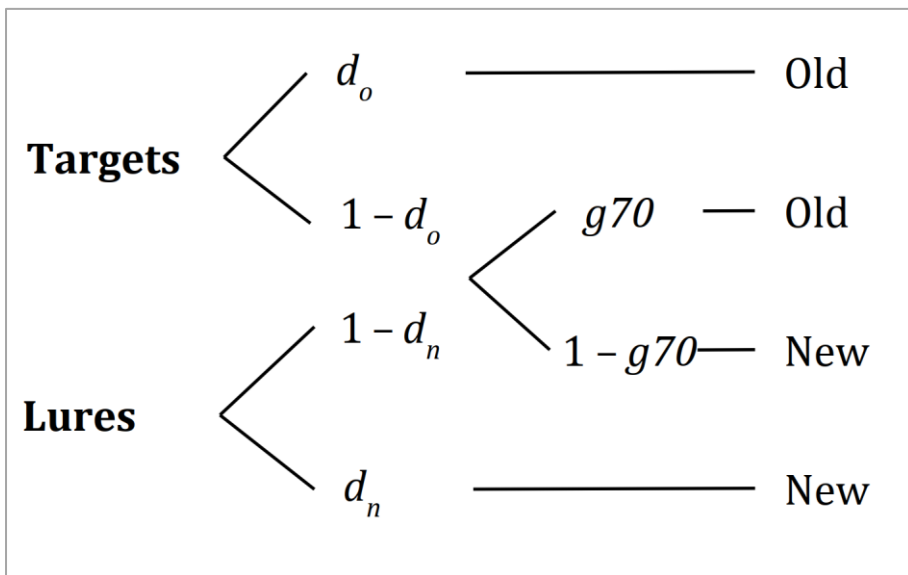
- Constraints: Assume  $d_o = d_n$
- More conditions: Base rate manipulation of response bias  $g$

# Identifiability: More Empirical Categories

(A) Base rate: 30% targets



(B) Base rate: 70% targets



→ Two additional degrees of freedom ( $df = 4$ )

→ But only *one* additional free parameter ( $S = 4$ )

→ The model is **identifiable**.

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# Parameter Estimation

- Given the data  $n_1, \dots, n_J$ , what is the „best“ vector of parameter values  $\theta = (\theta_1, \dots, \theta_s, \dots, \theta_S)$ 
  - Find  $\theta$  that minimizes the distance between observed and expected category frequencies!
- Distance measure: The likelihood ratio statistic  $G^2$

$$G^2(\theta) = -2 \sum_{j=1}^J n_j \ln \left( \frac{\overbrace{n_j}^{\text{observed frequencies}}}{\underbrace{N \cdot p_j(\theta)}_{\text{predicted frequencies}}} \right)$$

# Power Divergence Statistic

- Alternative distance measure: **Pearson  $X^2$**

$$X^2(\boldsymbol{\theta}) = \sum_{j=1}^J \frac{[n_j - N \cdot p_j(\boldsymbol{\theta})]^2}{N \cdot p_j(\boldsymbol{\theta})}$$

- Both types of distance measures ( $G^2$  and Pearson  $X^2$ ) are special cases of the **Power-Divergence-family** ( **$PD_\lambda$  statistics**) (Read & Cressie, 1988):

$$PD_\lambda = \frac{2}{\lambda(\lambda + 1)} \sum_{k=1}^K \sum_{i=1}^{J(k)} n_{j(k)} * \left[ \left( \frac{n_{j(k)}}{e_{j(k)}} \right)^\lambda - 1 \right]$$

**Note that:**

- Pearson  $X^2 = PD_{\lambda=1}$
- Likelihood-ratio statistic  $G^2 = \lim_{\lambda \rightarrow 0} PD_\lambda$

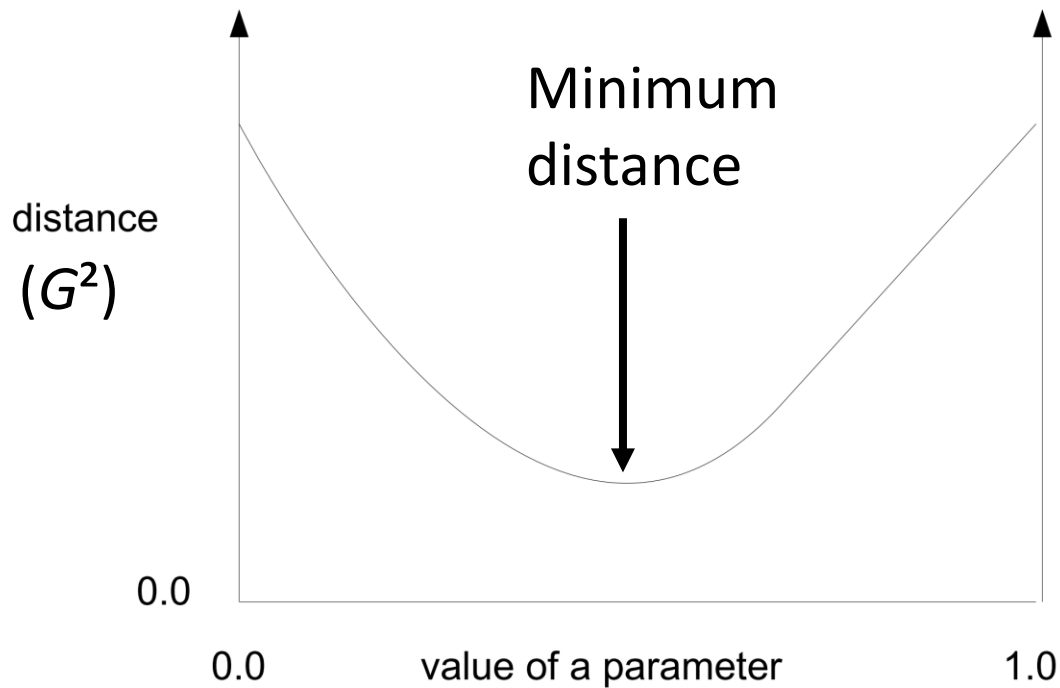
# What is the best goodness-of-fit statistic?

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- In case of **small sample sizes**,  $PD_{\lambda=1}$  and  $PD_{\lambda=2/3}$  outperform other  $PD_{\lambda}$ -statistics in terms of accuracy of  $\chi^2$  approximation (cf. Read & Cressie, 1988).
  - However, small samples are typically less of a problem in MPT model applications (unless models are tested for single participants)
- Given the fact that  $G^2$  is a by-product of ML-parameter estimation,  **$G^2 (= PD_{\lambda=0})$  can be recommended** for moderate to large sample sizes

# Parameter Estimation

- Which are the **best parameter values** given the data?
- Aim: Minimization of the distance measure  $G^2$
- Example: MPT model with  **$S = 1$  parameter**

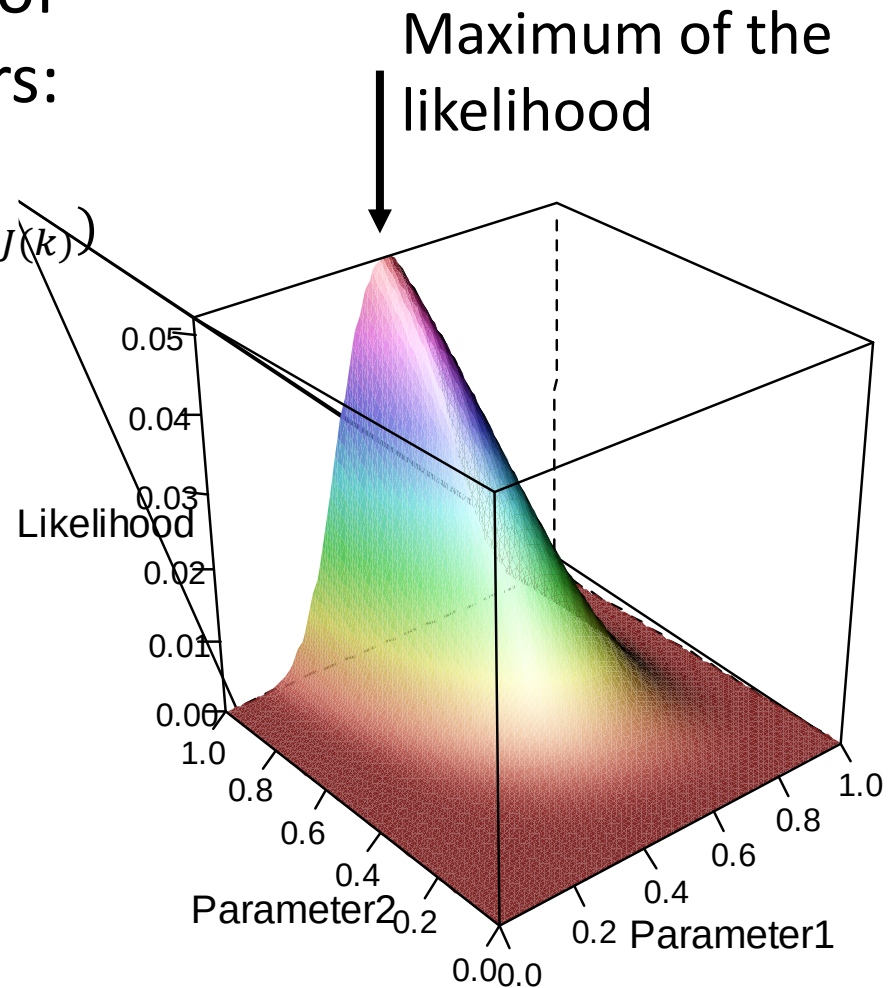


# Parameter Estimation

- Minimization of  $G^2$  is equivalent to **maximizing the likelihood** of the data given the parameters:

$$L(\theta ; n) = \prod_{k=1}^K p_{N(k), \pi(k)}(n_{1(k)}, n_{2(k)}, \dots, n_{J(k)})$$

- Example: MPT model with  **$S = 2$  parameters**





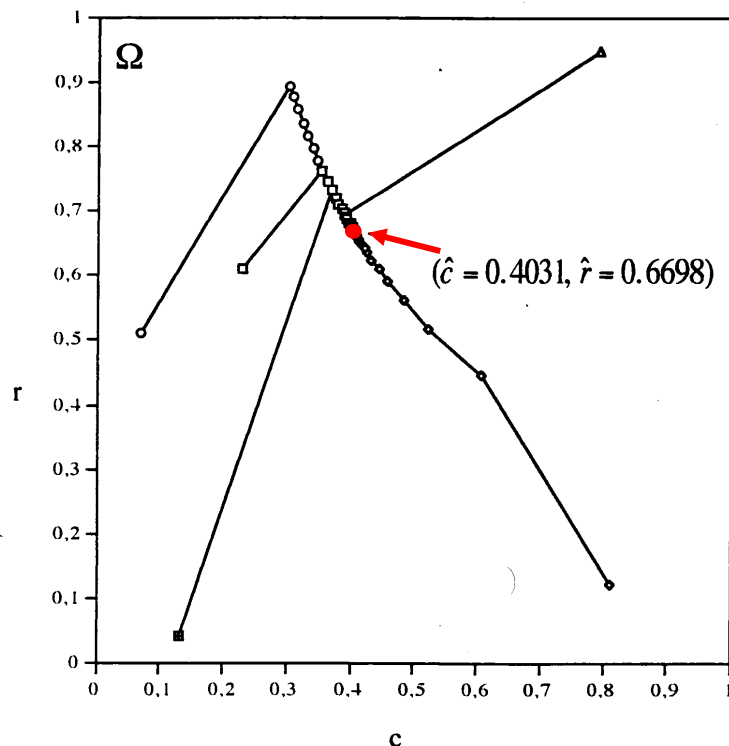
# Expectation-Maximization-(EM) Algorithm

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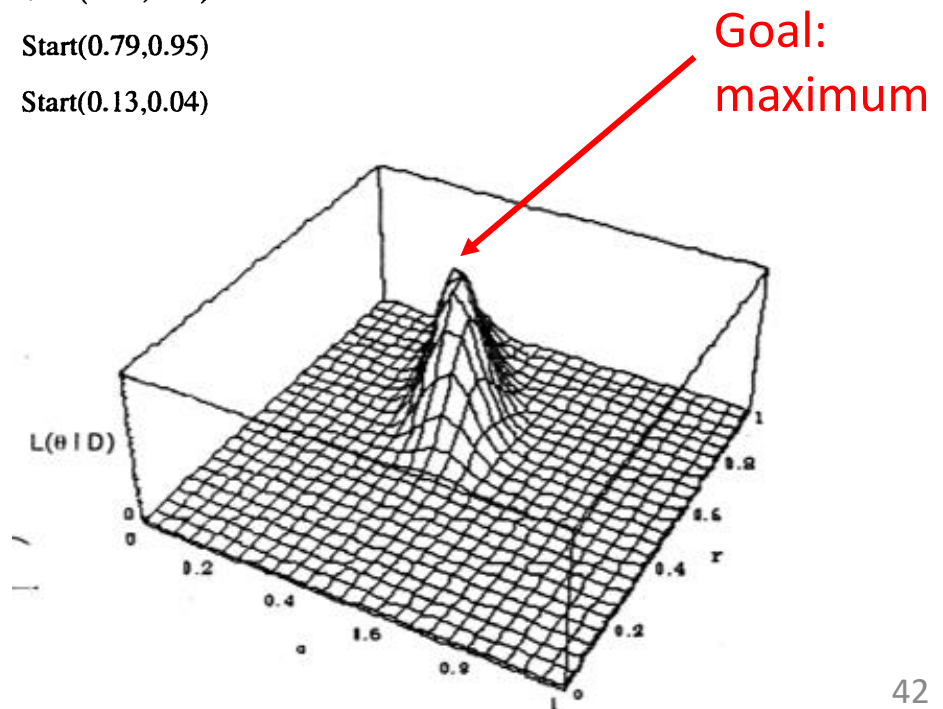
1. Choose a **random start vector**  $\theta_i$
2. **E(xpectation)-Step:**
  - Estimate the expected frequencies of the branches given  $\theta_i$  and the observed category frequencies  $n_{j(k)}$
3. **M(aximization)-Step:**
  - Let  $i = i + 1$
  - Compute new  $G^2$  estimates  $\theta_i$  given the expected frequencies from step 2
4. **Convergence?**
  - if  $|\theta_i - \theta_{i-1}| > \varepsilon \rightarrow$  go back to Step 2 ( $\varepsilon$  = convergence criterion)
  - otherwise  $\rightarrow$  accept  $\theta_i$  as final parameter estimates

# EM Algorithm

- Graphical illustration of the EM algorithm applied to an MPT model with  $S = 2$  parameters ( $c$  and  $r$ )
  - Different path of the EM algorithm for five starting values

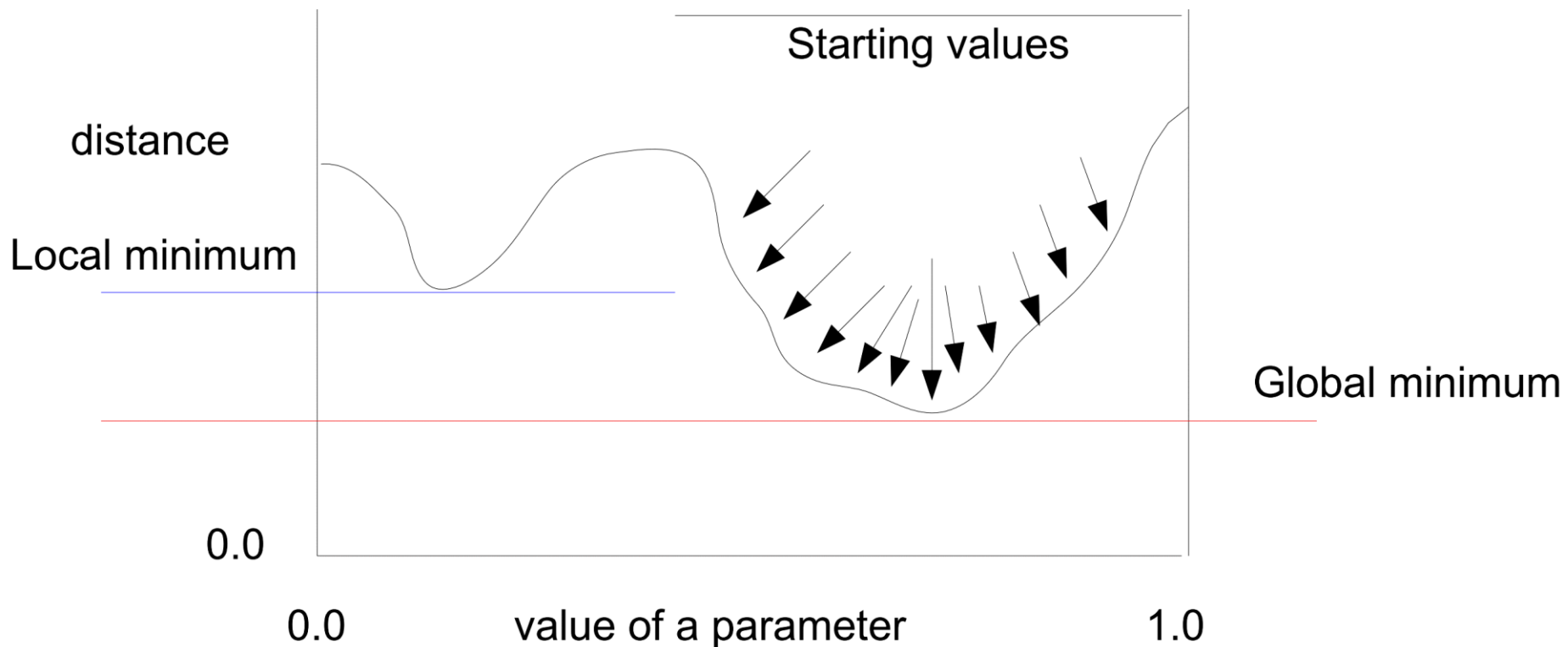


- Start(0.23,0.61)
- Start(0.81,0.12)
- Start(0.07,0.51)
- △— Start(0.79,0.95)
- Start(0.13,0.04)



# Parameter Estimation: Local Minima

- Possible issue: **Local minima** of the likelihood function
- Solution: **Fit model multiple times** with random starting values



# Basics of MPT Modeling

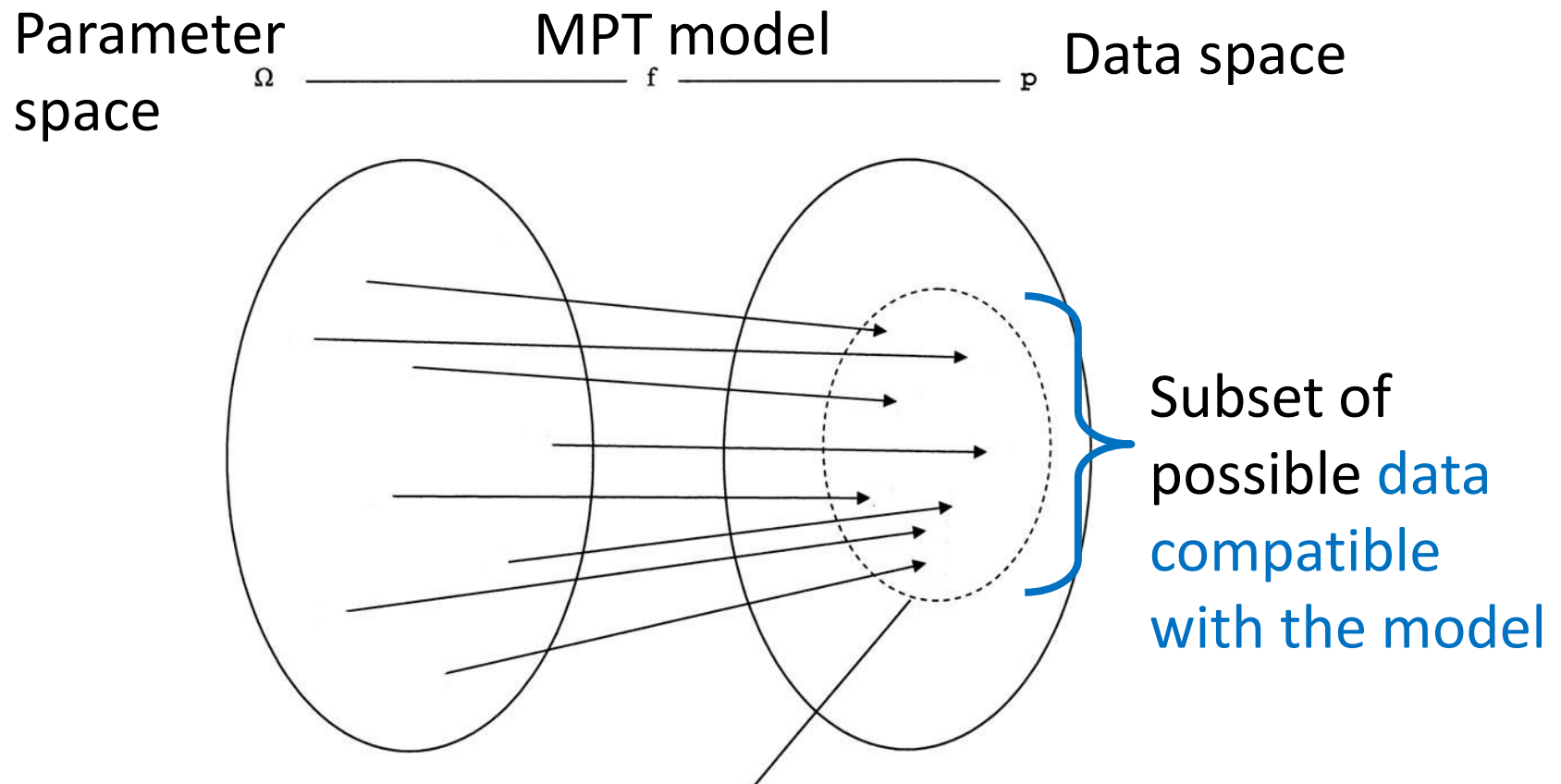
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# Model Assessment

- Graphical illustration of **model fit**:



# Model Assessment

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## Goodness-of-fit test

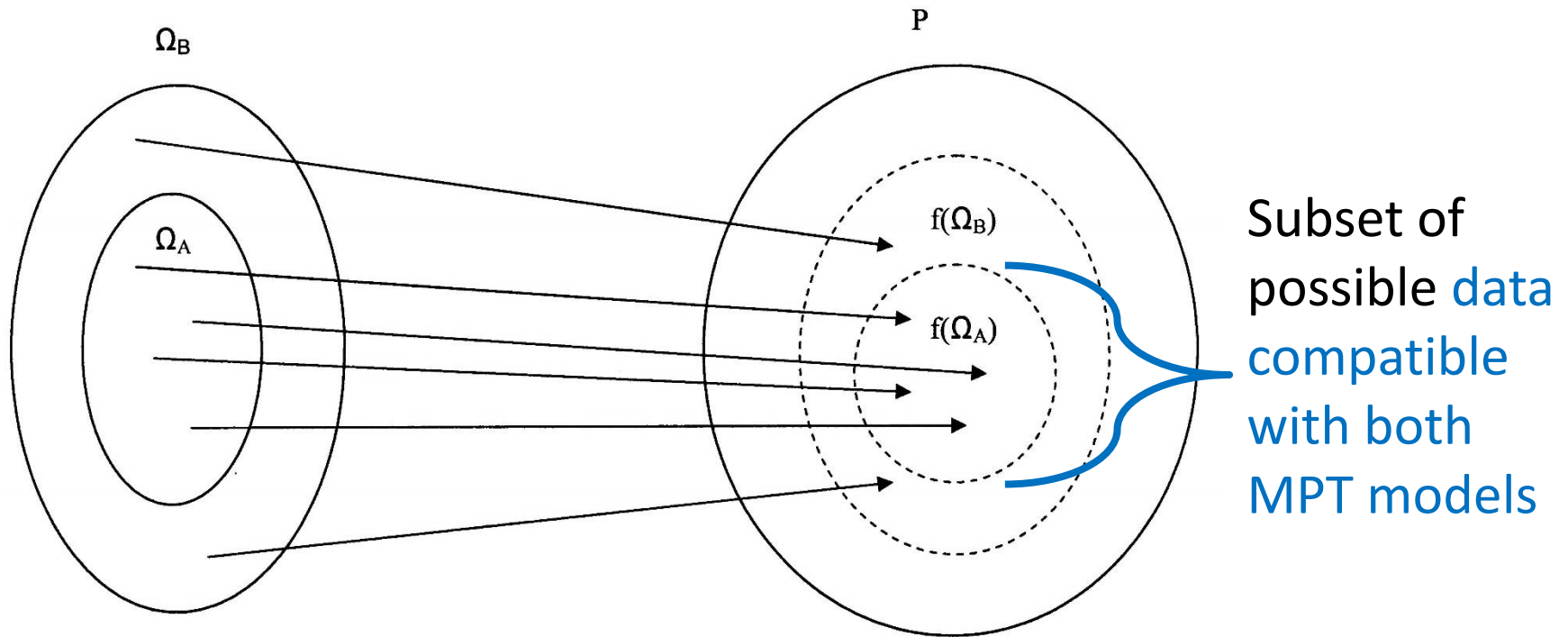
- How can we test **whether a model fits** the data?
- **Hypothesis**: the data are generated by the model
- $H_0: \pi \in f(\Omega)$ 
  - “the true category probabilities  $\pi$  are compatible with the model equations  $f(\Omega)$ ”
- Test statistic: Under  $H_0$ , the statistic  **$G^2$  is  $\chi^2$ -distributed** with degrees of freedom:

$$df = \sum_{k=1}^K (J_k - 1) - S$$

# Model Comparisons: Nested Models

## Hierarchical model families

- Model  $M_A$  is a **nested model** (= special case) of  $M_B$
- e.g., if  $M_A$  is obtained from  $M_B$  via **parameter restrictions**



# Model Comparisons: Hierarchical Model Families

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- If **model  $M_A$  is nested in  $M_B$**  and both are true then
  - $G^2_A$  is  $\chi^2$ -distributed with  $df_A$
  - $G^2_B$  is  $\chi^2$ -distributed with  $df_B$
  - $\Delta G^2_{A-B} = G^2_A - G^2_B$  is  $\chi^2$ -distributed  $df_{A-B} = df_A - df_B$
- Hence, we can use  $\Delta G^2_{A-B}$  to compare **nested models** using  $\chi^2$ -tests
- Unfortunately,  $\Delta G^2_{A-B}$  cannot be used for **non-nested models**
  - Solution: Information-theoretic measures (AIC, BIC)



# Model Selection: Information-Theoretic Measures

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- Core idea: Select the model that achieves the best tradeoff between **model fit vs. model complexity**
- Akaike Information Criterion (**AIC**)
  - $AIC(M_0) = -2 \log(L(\boldsymbol{\theta}; \mathbf{n})) + 2 S$
- Bayesian Information Criterion (**BIC**)
  - $BIC(M_0) = -2 \log(L(\boldsymbol{\theta}; \mathbf{n})) + S \log(N)$
- Application: Choose the model with the **smallest AIC/BIC**
- To assess model fit: Comparison to **saturated model**
  - $\Delta AIC(M_0) = AIC(M_0) - AIC(\text{saturated}) = G^2(M_0) - 2 \text{df}(M_0)$
  - $\Delta BIC(M_0) = BIC(M_0) - BIC(\text{saturated}) = G^2(M_0) - \text{df}(M_0) \log(N)$
  - Model fit is good if  **$\Delta AIC < 0$  or  $\Delta BIC < 0$**

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